

Lecture 14: Connections with Partial Differential Equations

October 27, 2010

1 Stochastic Differential Equations and Examples

We consider a stochastic differential equation of the form

$$dX(u) = \beta(u, X(u))du + \gamma(u, X(u))dW(u), \quad (1)$$

where $\beta(u, x)$ and $\gamma(u, x)$ are given deterministic functions, and $\{W(u)\}$ is a standard Brownian motion. At time $t > 0$ we fix a $x \in \mathbb{R}$, we can start a stochastic process $\{X(T)\}_{T \geq t}$ as

$$X(T) = x + \int_t^T \beta(u, X(u))du + \int_t^T \gamma(u, X(u))dW(u). \quad (2)$$

Under mild conditions on $\beta(u, x)$ and $\gamma(u, x)$, the equation (1) has a unique solution $\{X(T)\}$. But a general stochastic differential equation like (1) is hard to solve, since the unknown $X(u)$, $u > t$, appears in both sides of the equation. However, a one-dimensional linear stochastic differential equation can be explicit solved:

$$dX(u) = (\alpha(u) + b(u)X(u))du + (\gamma(u) + \sigma(u)X(u))dW(u), \quad (3)$$

where $a(u), b(u), \gamma(u)$ and $\sigma(u)$ are nonrandom functions of time. We study three examples below.

Example 1. The stochastic differential equation for a geometric Brownian motion is

$$dS(u) = \alpha S(u)du + \sigma S(u)dW(u). \quad (4)$$

If the initial value $S(t) = x > 0$ is given, then we have

$$S(T) = x \exp \left\{ \sigma (W(T) - W(t)) + \left(\alpha - \frac{1}{2} \sigma^2 \right) (T - t) \right\}. \quad (5)$$

Example 2. Hull White interest rate model is given by

$$dR(u) = (\alpha(u) - b(u)R(u))du + \sigma d\tilde{W}(u), \quad (6)$$

This model is a generalization of Vasicek interest rate model, in that it models the whole short rate curve. The way to solve it is similar as the Vasicek model. We work out the differential

$$d(e^{\int_0^u b(v)dv} R(u)) = e^{\int_0^u b(v)dv} (b(u)R(u)du + dR(u)) = e^{\int_0^u b(v)dv} (\alpha(u)du + \sigma(u)d\tilde{W}(u)). \quad (7)$$

Fix the initial condition $R(t) = r$, from we obtain that

$$e^{\int_0^T b(v)dv} R(T) = r e^{\int_0^t b(v)dv} + \int_t^T e^{\int_0^u b(v)dv} \alpha(u)du + \int_t^T e^{\int_0^u b(v)dv} \sigma(u)d\tilde{W}(u), \quad (8)$$

Finally we can solve for $R(T)$:

$$R(T) = r e^{-\int_t^T b(v)dv} + \int_t^T e^{-\int_u^T b(v)dv} \alpha(u)du + \int_t^T e^{-\int_u^T b(v)dv} \sigma(u)d\tilde{W}(u). \quad (9)$$

Notice that, $R(T)$ is normally distributed with mean

$$ER(T) = re^{-\int_t^T b(v)dv} + \int_t^T e^{-\int_u^T b(v)dv} a(u)du, \quad (10)$$

and variance

$$\text{Var}(R(T)) = \int_t^T e^{-\int_u^T b(v)dv} \sigma^2(u)du. \quad (11)$$

Example 3. Cox-Ingersoll-Ross interest rate model follows

$$dR(u) = \kappa(\theta - R(u))du + \sigma\sqrt{R(u)}d\tilde{W}(u). \quad (12)$$

This model does not admit a closed-form solution. But (12) do have a unique solution. The interest rate process $\{R(t)\}$ will not become negative.

2 The Markov Property

For all the above solutions to corresponding stochastic differential equations, we compute the conditional expectation

$$E(h(X(T))|\mathcal{F}_t), \text{ for a given function } h. \quad (13)$$

By definition of conditional expectation, the expression in (13) depends on the whole path $\{X(u)|u \in [0, t]\}$. But if it only depends on the value $X(t)$ and the time t , i.e., there exists a function $g(u, x)$ such that

$$E(h(X(T))|\mathcal{F}_t) = g(t, X(t)), \quad (14)$$

then we say the process $\{X(u)\}$ is a Markov process. Solutions to stochastic differential equations (with deterministic coefficients) are all Markov processes.

Markov processes facilitate the use of partial differential equations, as we will see in the next section.

3 Partial Differential Equations

Consider a stochastic differential equation

$$dX(u) = \beta(u, X(u))du + \sigma(u, X(u))dW(u). \quad (15)$$

We know that $\{X(u)\}$ is a Markov process, so for a function h , there exist a function $g(u, x)$ such that

$$g(t, x) = E(h(X(T))|X(t) = x). \quad (16)$$

Then clearly,

$$g(T, x) = x. \quad (17)$$

Moreover, for any $t \in (0, T)$, $g(t, x)$ satisfies a partial differential equation:

$$\frac{\partial}{\partial t}g(t, x) + \beta(t, x)\frac{\partial}{\partial x}g(t, x) + \frac{1}{2}\gamma^2(t, x)\frac{\partial^2}{\partial x^2}g(t, x) = 0. \quad (18)$$

The above result is called the **Feynmann-Kac theorem**. It suggests a PDE approach to obtain the conditional expectation for solutions of stochastic differential equations.

Why the result is true? Because the conditional expectation $\{E(h(X(T))|\mathcal{F}_t)\}_{t \in [0, T]}$ defines a martingale. As a result, the process $E(h(X(T))|\mathcal{F}_t) = g(t, X(t))$ should have zero drift. We apply Itô's formula to $g(t, X(t))$, then we have that

$$\begin{aligned} dg(t, X(t)) &= g_t dt + g_x dX(t) + \frac{1}{2}g_{xx}(dX(t))^2 = g_t dt + \beta g_x dt + \gamma g_x dW(t) + \frac{1}{2}\gamma^2 g_{xx} dt \\ &= \left(g_t(t, X(t)) + \beta(t, X(t))g_x(t, X(t)) + \frac{1}{2}\gamma^2(t, X(t))g_{xx}(t, X(t)) \right) dt + \gamma(t, X(t))g_x(t, X(t))dW(t) \end{aligned} \quad (19)$$

So we must have (18).

In derivative pricing, we usually need to know the discounted expected payoff, instead of the payoff itself. More specifically, we would like to solve for a function $g(u, x)$, such that

$$g(t, x) = E(e^{-r(T-t)}h(X(T))|X(t) = x), \quad t \in [0, T]. \quad (20)$$

If so, the process $\{e^{-rt}g(t, X(t))\}_{t \in [0, T]}$ will be a martingale¹, which has no drift. Itô's formula tell us

$$\begin{aligned} d(e^{-rt}g(t, X(t))) &= e^{-rt} \left(-gdt + g_t dt + f_x dX(t) + \frac{1}{2}g_{xx}(dX(t))^2 \right) \\ &= e^{-rt} \left(-rg + g_t + \beta g_x + \frac{1}{2}\gamma^2 g_{xx} \right) dt + e^{-rt}\gamma g_x dW(t). \end{aligned} \quad (21)$$

So we must have

$$\frac{\partial}{\partial t}g(t, x) + \beta(t, x)\frac{\partial}{\partial x}g(t, x) + \frac{1}{2}\gamma^2(t, x)\frac{\partial^2}{\partial x^2}g(t, x) = rg(t, x). \quad (22)$$

4 Application in Risk-neutral Pricing

Consider a complete arbitrage-free market (so risk-neutral pricing is justified), we assume that, under the risk-neutral measure $\tilde{p}$, the asset price process follows

$$dS(u) = rS(u) + \sigma(t, S(u))S(u)d\tilde{W}(u), \quad S(0) > 0. \quad (23)$$

Then any European contract with payoff $h(S(T))$ can be valued by solve the generalized Black-Scholes equation:

$$\frac{\partial}{\partial t}g(t, x) + rx\frac{\partial}{\partial x}g(t, x) + \frac{1}{2}\sigma^2(t, x)x^2\frac{\partial^2}{\partial x^2}g(t, x) = rg(t, x). \quad (24)$$

This equation can be solved numerically. And if we are at time t , the value of the option is simply $g(t, S(t))$.

If we choose constant volatility $\sigma(t, x) = \sigma > 0$, we have classical Black-Scholes model. However, the model cannot produce the price seen in the market. It is seen that, to match the market prices, we have to use different volatilities. The volatility that can produce the market price is called the implied volatility. For call options with the same expiration, the implied volatility is a convex function of the strike price. This is the so-called volatility smile. For options with different maturities, the shape of volatility smile are different. So finally, the implied volatility depends on both the strike price and the maturity, this gives us the so-called volatility surface. Jim Gatheral has a good book on it!

¹This is because, $E(e^{-rT}h(X(T))|\mathcal{F}_t) = e^{-rt}g(t, X(t))$ is a martingale.